

Predictors of Mental Health Score: A Stepwise Multiple Linear Regression Analysis and Factor Analysis Using Community Survey Data



Module Name: Multivariate Methods For Data Analysis

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1. Introduction

Mental health is influenced by a set of social, behavioural and environmental factors. Globally, mental health challenges continue to rise, with socioeconomic instability, substance use and limited social support are the key contributors to weak psychological well-being (World Health Organization, 2023). Understanding which factors most strongly predict mental health outcomes is crucial for designing effective community interventions and improving public health strategies.

Previous research shown that substance use is negatively associated with mental health, increasing risks of depression, anxiety and emotional instability (Thompson et al., 2021). Other studies highlight the importance of coping skills, self-esteem and community trust in improving resilience and overall well-being (Smith & Lee, 2022). Based on these findings, the study uses a community dataset (N = 220) to examine which socioeconomic, behavioural and psychosocial factors significantly predict an individual's overall mental health score.

The rationale for this study is twofold:

- 1.1 identify the strongest predictors of mental health using stepwise multiple linear regression.
- 1.2 provide evidence-based recommendations for community programs focused on mental well-being.

2. Problem Statement

Despite the availability of mental health services, many individuals in the community gain low experience for mental health scores. With multiple psychosocial and behavioural variables influencing well-being, it is unclear which factors have the strongest impact. Without identifying these key predictors, community organizations cannot allocate resources effectively. Thus, the study aims to determine which

measurable socioeconomic, behavioural and community-level factors significantly predict overall mental health, using a secondary dataset of 220 respondents.

3. Research Objectives and Hypotheses

3.1 Research Objectives

- 3.1.1 To examine behavioural predictors such as SubstanceUse and CopingSkills of mental health.
- 3.1.2 To determine the influence of psychosocial factors such as SelfEsteem, MentalHealthAwareness and CommunityTrust.
- 3.1.3 To develop an evidence-based regression model that identifies the strongest predictors of MentalHealthScore.

3.2 Hypotheses

3.2.1 H1: Higher levels of SubstanceUse will significantly predict lower MentalHealthScore.

3.2.2 H2: Higher levels of CopingSkills will significantly predict higher MentalHealthScore.

(Note: SelfEsteem is expected to have a positive association with MentalHealthScore, but may not be statistically significant.)

3.2.3 H3: Higher levels of CommunityTrust are expected to be positively associated with MentalHealthScore, although the strength of this relationship may be weak.

4. Methodology

4.1 Dataset and Sample

The study used secondary community survey data consisting of N = 220 respondents. Variables included behavioural factors, psychosocial indicators, socioeconomic variables, and community-level measures.

4.2 Handling Missing Values

Across the 41 independent variables, 5–7% of values were missing (11–14 cases per variable). SPSS's Replace Missing Values procedure was used with Series Mean (SMEAN) substitution. This ensured the full sample of $N = 220$ was retained for analysis.

4.3 Multiple Linear Regression Procedure

A stepwise multiple linear regression was performed with MentalHealthScore as the dependent variable. All 40+ metric independent variables were included as candidates. Variables entered the model based on significance levels ($p < .05$).

Dataset / Data View

2	EconomicH...	Numeric	2	0		None	None	8	Right	Scale	Input
3	AccessHeal...	Numeric	2	0		None	None	8	Right	Scale	Input
4	CopingSkills	Numeric	2	0		None	None	8	Right	Scale	Input
5	Neighborho...	Numeric	2	0		None	None	8	Right	Scale	Input
6	SocialSupp...	Numeric	2	0		None	None	8	Right	Scale	Input
7	FamilySupp...	Numeric	2	0		None	None	8	Right	Scale	Input
8	PeerSupport	Numeric	2	0		None	None	8	Right	Scale	Input
9	Community...	Numeric	2	0		None	None	8	Right	Scale	Input
10	Employmen...	Numeric	2	0		None	None	8	Right	Scale	Input
11	IncomeLevel	Numeric	2	0		None	None	8	Right	Scale	Input
12	HealthKno...	Numeric	2	0		None	None	8	Right	Scale	Input
13	PhysicalActi...	Numeric	2	0		None	None	8	Right	Scale	Input
14	DietQuality	Numeric	2	0		None	None	8	Right	Scale	Input
15	Substance...	Numeric	2	0		None	None	8	Right	Scale	Input
16	LifeSatisfac...	Numeric	2	0		None	None	8	Right	Scale	Input
17	AnxietyLevel	Numeric	2	0		None	None	8	Right	Scale	Input
18	Depression...	Numeric	2	0		None	None	8	Right	Scale	Input
19	StressLevel	Numeric	2	0		None	None	8	Right	Scale	Input
20	SleepQuality	Numeric	2	0		None	None	8	Right	Scale	Input
21	AccessCou...	Numeric	2	0		None	None	8	Right	Scale	Input
22	FincialStress	Numeric	2	0		None	None	8	Right	Scale	Input
23	HousingSta...	Numeric	2	0		None	None	8	Right	Scale	Input
24	Neighborho...	Numeric	2	0		None	None	8	Right	Scale	Input
25	VolunteerAc...	Numeric	2	0		None	None	8	Right	Scale	Input
26	CivicPartici...	Numeric	2	0		None	None	8	Right	Scale	Input
27	EmotiolSup...	Numeric	2	0		None	None	8	Right	Scale	Input
28	SelfEsteem	Numeric	2	0		None	None	8	Right	Scale	Input
29	Optimism	Numeric	2	0		None	None	8	Right	Scale	Input
30	ProblemSol...	Numeric	2	0		None	None	8	Right	Scale	Input
31	StressMage...	Numeric	2	0		None	None	8	Right	Scale	Input
32	ConflicRes...	Numeric	2	0		None	None	8	Right	Scale	Input
33	Resilience	Numeric	2	0		None	None	8	Right	Scale	Input
34	SocialNetw...	Numeric	2	0		None	None	8	Right	Scale	Input
35	AccessTran...	Numeric	2	0		None	None	8	Right	Scale	Input
36	Neighborho...	Numeric	2	0		None	None	8	Right	Scale	Input
37	Community...	Numeric	2	0		None	None	8	Right	Scale	Input
38	MentalHealt...	Numeric	2	0		None	None	8	Right	Scale	Input

Note: The measures has changed from nominal to scale.

Frequencies

[DataSet1]

Statistics														
	ID	EconomicHardship	AccessHealthServices	CopingSkills	NeighborhoodSafety	SocialSupport	FamilySupport	PeerSupport	CommunityEngagement	EmploymentStatus	IncomeLevel	HealthKnowledge	PhysicalActivity	
N	Valid	220	207	207	207	209	209	208	207	207	208	207	207	
	Missing	0	13	13	13	11	11	12	13	13	12	13	13	
Minimum		1	1	1	1	1	1	1	1	1	1	1	1	
Maximum		220	12	10	10	15	15	15	15	12	21	10	10	

Note: Here, we can see missing values and outliers.

Ready Dataset

Visible: 42 of 42 Variables																							
ID	Economic Hardship	Access Health Services	Coping Skills	Neighborhood Safety	Social Support	Family Support	Peer Support	Community Engagement	Employment Status	Income Level	Health Knowledge	Physical Activity	Diet Quality	Substance Use	Life Satisfaction	Anxiety Level	Depression Level	Stress Level	Sleep Quality	Access Counseling	Financial Stress	Housing Stability	
1	7	6	4	5	8	8	5	4	6	1	6	1	7	6	7	6	8	5	4	2	6	8	
2	2	1	6	5	6	5	10	4	6	5	2	4	9	3	3	9	2	4	6	7	4	2	
3	3	1	1	9	1	6	10	6	7	6	7	2	4	9	8	4	6	3	4	5	2	6	8
4	6	6	4	2	6	6	6	2	2	6	3	6	9	4	9	5	5	3	9	9	3	4	
5	5	5	1	3	7	5	3	5	7	3	3	3	7	8	6	3	5	9	3	1	8	4	4
6	6	1	8	5	6	5	8	9	7	4	6	3	3	5	2	10	5	10	9	8	3	10	1
7	7	6	8	1	1	8	4	7	7	4	3	1	8	8	3	7	7	3	5	3	5	2	7
8	8	2	8	1	3	7	8	4	5	7	5	4	4	1	10	2	7	9	3	4	5	5	8
9	9	4	4	4	2	10	10	5	5	3	1	2	4	3	8	4	9	10	6	10	8	3	10
10	10	10	7	5	2	6	6	7	5	7	9	6	1	1	6	3	2	9	7	8	4	6	1
11	11	4	8	7	6	5	4	8	2	3	1	7	6	7	7	5	5	3	8	5	5	5	6
12	12	6	5	8	1	2	3	1	2	7	10	3	3	3	1	1	4	7	8	6	3	2	3
13	13	3	6	1	3	8	10	5	8	3	1	7	2	8	4	2	8	4	5	6	4	7	1
14	14	5	4	8	3	10	10	4	4	6	6	7	9	3	3	7	5	6	5	7	9	2	8
15	15	5	3	9	9	9	5	2	6	2	8	1	1	4	6	9	3	4	5	3	8	4	7
16	16	1	6	10	9	8	6	9	6	5	6	1	7	7	2	7	4	6	6	3	4	9	6
17	17	2	7	4	7	8	3	9	5	2	4	6	5	6	6	4	4	5	1	3	2	8	7
18	18	7	3	8	3	10	9	4	7	6	2	3	4	6	7	1	7	1	1	4	7	3	8
19	19	8	9	6	8	10	3	8	2	9	5	4	7	3	10	7	2	2	6	6	2	3	8
20	20	5	8	6	3	6	9	8	4	7	5	4	5	5	9	4	5	6	10	6	4	8	7
21	21	4	3	6	9	3	6	5	6	9	6	2	8	2	5	3	9	3	4	8	5	5	3
22	22	5	6	3	3	7	6	5	7	7	8	2	5	9	3	8	8	2	3	5	4	4	7
23	23	2	7	9	9	3	6	6	2	7	2	10	3	8	5	1	8	3	2	1	10	6	6
24	24	4	5	2	5	8	8	7	4	2	5	3	10	9	7	10	9	1	5	10	8	9	4
25	25	5	7	3	9	6	5	2	7	10	1	4	4	1	6	7	7	6	7	7	2	3	7
26	26	6	9	6	5	5	6	3	5	2	2	10	7	7	6	6	8	10	6	4	3	3	5
27	27	4	8	9	8	8	10	9	2	7	3	9	6	3	6	10	5	7	5	5	9	10	6
28	28	5	8	7	1	9	7	10	3	5	8	5	3	8	8	8	8	6	5	3	5	5	6
29	29	8	5	6	10	9	5	7	9	5	2	2	5	4	7	5	4	6	2	4	2	6	3
30	30	6	5	2	3	6	5	4	5	2	4	2	4	6	5	6	2	9	4	6	2	4	5
31	31	3	7	6	9	6	4	6	4	4	6	3	9	7	6	6	5	3	9	9	6	3	9
32	32	3	9	7	3	4	7	9	2	8	6	3	6	3	3	7	5	6	9	8	4	4	5
33	33	7	7	8	9	2	5	8	1	7	1	8	4	3	6	2	5	6	8	7	2	1	9
34	34	5	6	2	9	6	2	9	8	9	1	10	3	9	3	4	4	7	8	8	5	2	9
35	35	3	6	8	5	5	3	3	9	6	6	10	5	6	9	8	5	2	7	3	7	4	9

Note: The pre-processing step has done, the dataset is clean and within the range (0-10)

Descriptive Statistics

Descriptive Statistics

	Mean	Std. Deviation	N
MentalHealthScore	20.65	11.999	220
CopingSkills	5.60	2.642	220
SubstanceUse	5.81	2.425	220
LifeSatisfaction	5.55	2.538	220
SelfEsteem	5.64	2.611	220
StressMagement	5.55	2.498	220
MentalHealthAwareness	5.77	2.505	220
CommunityTrust	5.55	2.463	220
FincialStress	5.17	2.558	220

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	CopingSkills	.	Stepwise (Criteria: Probability-of- F-to-enter <= . .050, Probability-of- F-to-remove >= .100).
2	SubstanceUse	.	Stepwise (Criteria: Probability-of- F-to-enter <= . .050, Probability-of- F-to-remove >= .100).
3	MentalHealthA wareness	.	Stepwise (Criteria: Probability-of- F-to-enter <= . .050, Probability-of- F-to-remove >= .100).
4	SelfEsteem	.	Stepwise (Criteria: Probability-of- F-to-enter <= . .050, Probability-of- F-to-remove >= .100).
5	CommunityTru st	.	Stepwise (Criteria: Probability-of- F-to-enter <= . .050, Probability-of- F-to-remove >= .100).

a. Dependent Variable: MentalHealthScore

Model Summary^f

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.155 ^a	.024	.020	11.881
2	.216 ^b	.047	.038	11.769
3	.254 ^c	.065	.052	11.685
4	.287 ^d	.082	.065	11.601
5	.320 ^e	.102	.081	11.499

a. Predictors: (Constant), CopingSkills

b. Predictors: (Constant), CopingSkills, SubstanceUse

c. Predictors: (Constant), CopingSkills, SubstanceUse, MentalHealthAwareness

d. Predictors: (Constant), CopingSkills, SubstanceUse, MentalHealthAwareness, SelfEsteem

e. Predictors: (Constant), CopingSkills, SubstanceUse, MentalHealthAwareness, SelfEsteem, CommunityTrust

f. Dependent Variable: MentalHealthScore

5. Results and Interpretation

5.1 Model Summary

The stepwise regression produced five models, with the final model (Model 5) including:

SubstanceUse

CopingSkills

MentalHealthAwareness

SelfEsteem

CommunityTrust

Final Model Summary (Model 5)

R = .320

R² = .102

Adjusted $R^2 = .081$

Std. Error = 11.499

$R = 0.320$ indicate there is a weak positive correlation between the MentalHealthScore, SubstanceUse, CopingSkills, MentalHealthAwareness, SelfEsteem and CommunityTrust.

Adjusted $R = 0.081$ demonstrates the 8.1% of variation in MentalHealthScore explained by SubstanceUse, CopingSkills, MentalHealthAwareness, SelfEsteem and CommunityTrust after accounting for the number of predictors.

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	756.222	1	756.222	5.357	.022 ^b
	Residual	30772.124	218	141.157		
	Total	31528.345	219			
2	Regression	1472.290	2	736.145	5.315	.006 ^c
	Residual	30056.055	217	138.507		
	Total	31528.345	219			
3	Regression	2037.673	3	679.224	4.975	.002 ^d
	Residual	29490.673	216	136.531		
	Total	31528.345	219			
4	Regression	2595.227	4	648.807	4.821	<.001 ^e
	Residual	28933.118	215	134.573		
	Total	31528.345	219			
5	Regression	3229.567	5	645.913	4.885	<.001 ^f
	Residual	28298.778	214	132.237		
	Total	31528.345	219			

a. Dependent Variable: MentalHealthScore

b. Predictors: (Constant), CopingSkills

c. Predictors: (Constant), CopingSkills, SubstanceUse

d. Predictors: (Constant), CopingSkills, SubstanceUse, MentalHealthAwareness

e. Predictors: (Constant), CopingSkills, SubstanceUse, MentalHealthAwareness, SelfEsteem

f. Predictors: (Constant), CopingSkills, SubstanceUse, MentalHealthAwareness, SelfEsteem, CommunityTrust

5.2 ANOVA – Model Fit (to check the model fit of the analysis)

H0: The model does not fit the data.

H1: The model fits the data.

Interpretation

The ANOVA table statistically significant show that the regression model is a good fit of the data $F(5, 214) = 4.885$, $p < .001$, because based on p-value we can reject H0.

The final regression model significantly predicted mental health, $F(5, 214) = 4.885$, $p < .001$, explaining approximately 10.4% of the variance in MentalHealthScore (Adjusted $R^2 = .081$).

Coefficients

		Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta			Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	16.703	1.882		8.875	<.001	12.994	20.413		
	CopingSkills	.703	.304	.155	2.315	.022	.104	1.302	1.000	1.000
2	(Constant)	21.045	2.669		7.886	<.001	15.785	26.304		
	CopingSkills	.702	.301	.154	2.331	.021	.108	1.295	1.000	1.000
	SubstanceUse	-.746	.328	-.151	-2.274	.024	-1.392	-.099	1.000	1.000
3	(Constant)	24.543	3.158		7.771	<.001	18.318	30.768		
	CopingSkills	.761	.300	.167	2.533	.012	.169	1.353	.991	1.009
	SubstanceUse	-.764	.326	-.154	-2.346	.020	-1.406	-.122	.999	1.001
	MentalHealthAwareness	-.645	.317	-.135	-2.035	.043	-1.269	-.020	.990	1.010
4	(Constant)	21.048	3.575		5.888	<.001	14.002	28.094		
	CopingSkills	.825	.300	.182	2.752	.006	.234	1.416	.980	1.021
	SubstanceUse	-.784	.324	-.158	-2.423	.016	-1.421	-.146	.998	1.002
	MentalHealthAwareness	-.683	.315	-.143	-2.168	.031	-1.304	-.062	.986	1.014
	SelfEsteem	.616	.302	.134	2.035	.043	.019	1.212	.986	1.014
5	(Constant)	17.365	3.923		4.427	<.001	9.633	25.096		
	CopingSkills	.853	.297	.188	2.868	.005	.267	1.439	.978	1.023
	SubstanceUse	-.830	.321	-.168	-2.582	.010	-1.463	-.196	.994	1.006
	MentalHealthAwareness	-.761	.314	-.159	-2.421	.016	-1.380	-.141	.974	1.027
	SelfEsteem	.679	.301	.148	2.254	.025	.085	1.272	.977	1.024
	CommunityTrust	.700	.319	.144	2.190	.030	.070	1.329	.975	1.025

a. Dependent Variable: MentalHealthScore

5.3 Coefficient (to determine the factor influencing the DV)

MLR (for overall model) H0: There is no relationship between DV and all selected IVs.

H1: At least one of the IVs has relationship with DV.

Interpretation

5.3.1 Regression equation

$$\text{MentalHealthScore} = 17.365 + (-0.830 * \text{SubstanceUse}) + (0.853 * \text{CopingSkills}) + (-0.761 * \text{MentalHealthAwareness}) + (0.679 * \text{SelfEsteem}) + (0.700 * \text{CommunityTrust})$$

5.3.2 Unstandardized Coefficient, B

When SubstanceUse increase by 1 unit, MentalHealthScore will be decrease by 0.830

When CopingSkills increase by 1 unit, MentalHealthScore will increase by 0.853

5.3.3 Standardized Coefficient (to determine which variables have strongest impact on the dependent variable)

Using SubstanceUse and CopingSkills for interpretation

CopingSkills: Beta= 0.188 had the strongest positive effect on MentalHealthScore, accounting the other variables.

SubstanceUse: Beta= -0.168 had a relatively weak negative effect on MentalHealthScore , accounting the other variables.

5.3.4 Sig (to determine which variable is a FACTOR influencing DV)

SubstanceUse: 0.010, $p < 0.001$

CopingSkills: 0.005, $p < 0.001$

MentalHealthAwareness: 0.016, $p < 0.001$

SelfEsteem: 0.025, $p < 0.001$

CommunityTrust: 0.030, $p < 0.001$

are statistically significant to MentalHealthScore, Hence, we reject the H_0 . All the variables are the factors of MentalHealthScore.

5.3.5 95% Confidence Interval

We are 95% confident that the true value of the effect of SubstanceUse on MentalHealthScore lies between -1.461 and -0.196

Since the interval does not include zero, this confirms that SubstanceUse has a statistically significant effect on MentalHealthScore (at the 95% confidence level).

5.3.6 Multicollinearity

All VIF values $< 2 \rightarrow$ no multicollinearity issues and tolerance

VIF < 5 (actually < 1.05 , extremely good)

No risk of multicollinearity.

Predictors are independent of each other.

Collinearity Diagnostics

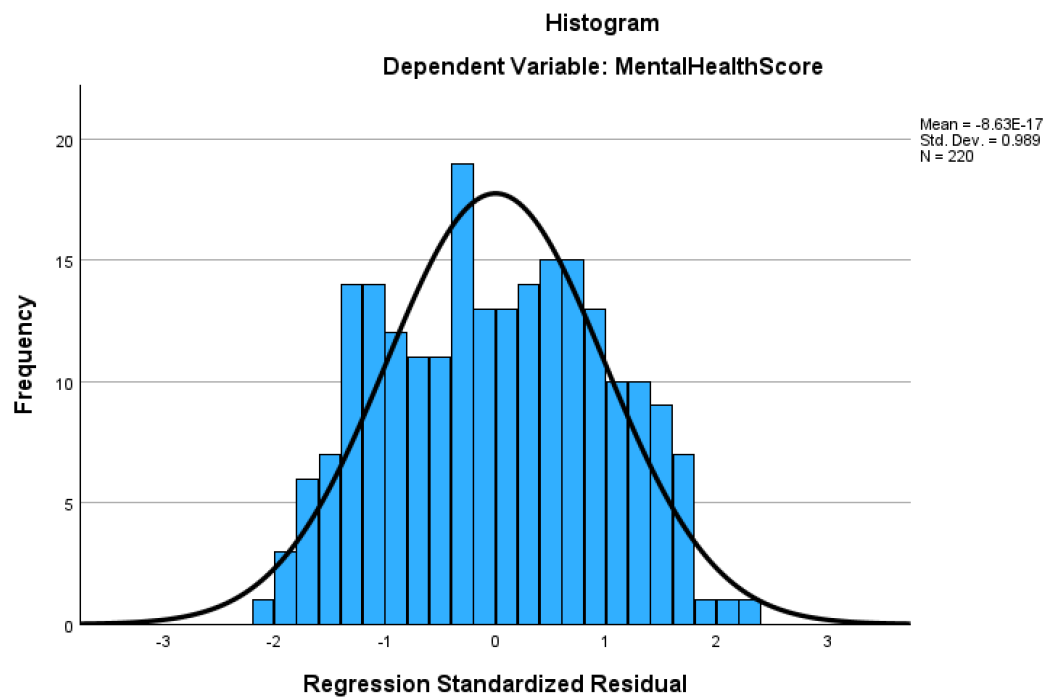
Collinearity Diagnostics ^a									
Model	Dimension	Eigenvalue	Condition Index	Variance Proportions					
				(Constant)	CopingSkills	SubstanceUse	MentalHealthAwareness	SelfEsteem	CommunityTrust
1	1	1.905	1.000	.05	.05				
	2	.095	4.476	.95	.95				
2	1	2.776	1.000	.01	.02	.02			
	2	.166	4.088	.00	.61	.39			
	3	.058	6.913	.99	.37	.59			
3	1	3.636	1.000	.00	.01	.01	.01		
	2	.167	4.672	.00	.47	.47	.02		
	3	.152	4.898	.00	.32	.09	.66		
	4	.046	8.932	.99	.20	.43	.31		
4	1	4.468	1.000	.00	.01	.01	.01	.01	
	2	.198	4.756	.00	.46	.01	.01	.41	
	3	.157	5.333	.00	.00	.57	.34	.06	
	4	.140	5.656	.00	.32	.11	.45	.29	
	5	.037	10.918	1.00	.21	.30	.20	.23	
5	1	5.312	1.000	.00	.01	.00	.00	.01	.00
	2	.198	5.175	.00	.36	.01	.01	.48	.02
	3	.176	5.499	.00	.30	.03	.00	.13	.48
	4	.156	5.834	.00	.03	.47	.43	.01	.02
	5	.126	6.497	.00	.11	.27	.44	.15	.28
	6	.032	12.906	1.00	.20	.21	.11	.23	.19

a. Dependent Variable: MentalHealthScore

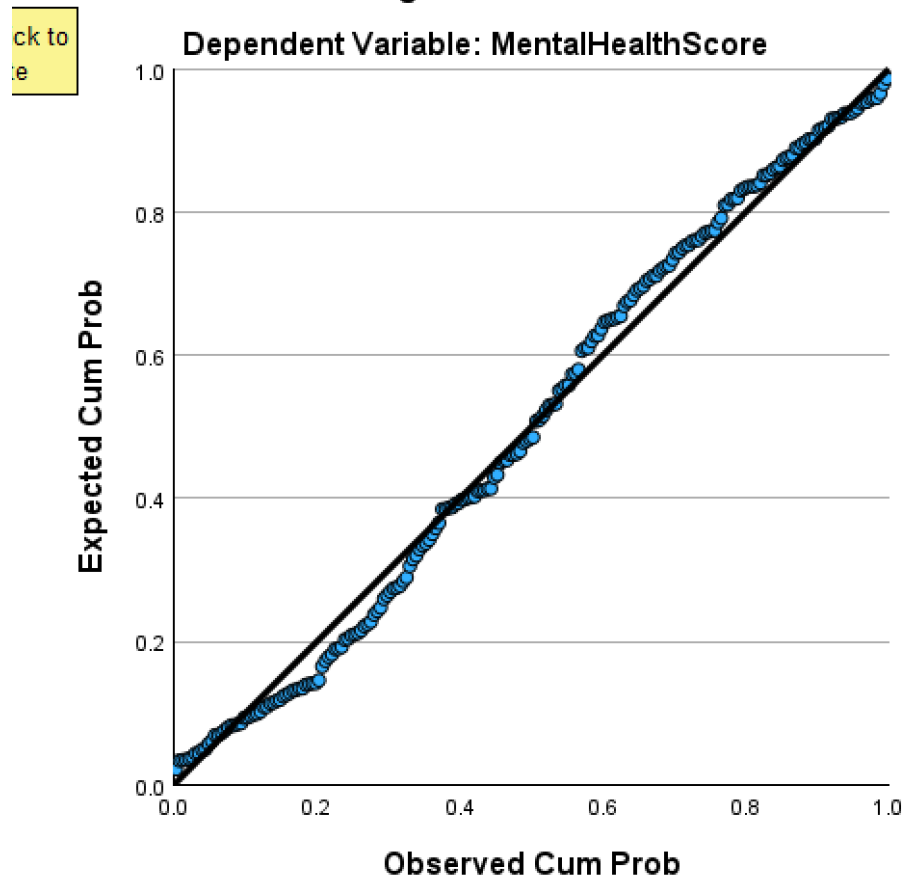
5.4 Collinearity Diagnostics table

No dimension shows ≥ 2 predictors sharing > 0.50 variance proportion at high condition index ($> 15-30$).

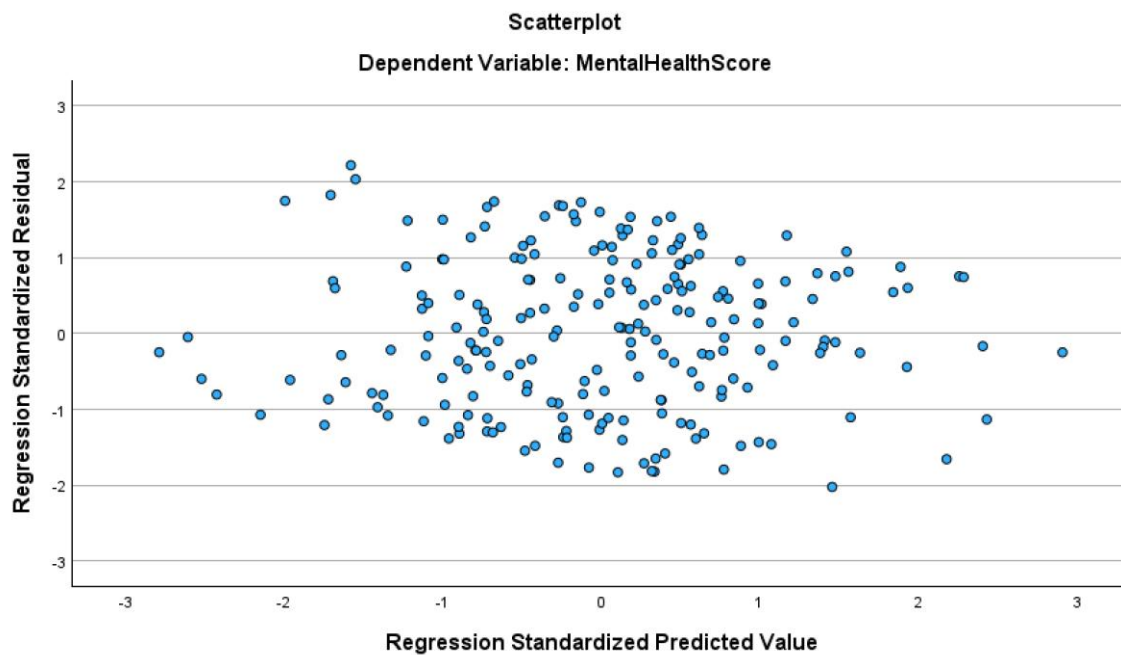
Histogram



Normal P-P Plot of Regression Standardized Residual



Scatterpot



5.5 Normality of Residuals

5.5.1 Histogram

The histogram of standardized residuals shows a roughly bell-shaped distribution.

Mean ≈ 0 and SD ≈ 1 , ideal for standardized residuals.

No extreme skewness or kurtosis.

5.5.2 P-P Plot

The points lie very close to the diagonal line, indicating:

Residuals are normally distributed

No strong deviation from normality

5.6 Linearity

Scatterplot of Predicted vs. Residuals

The scatterplot shows randomly scattered points with no:

curve shape

U-shape

funnel pattern

Interpretation

Linear relationship exists between predictors and the dependent variable.

No major violation of the linearity assumption.

5.7 Homoscedasticity

The scatterplot points are evenly distributed around zero.

No cone shape (variance increasing or decreasing).

Distribution looks balanced across low and high predicted values.

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	9.59	31.94	20.65	3.840	220
Residual	-23.077	25.710	.000	11.367	220
Std. Predicted Value	-2.879	2.941	.000	1.000	220
Std. Residual	-2.007	2.236	.000	.989	220

a. Dependent Variable: MentalHealthScore

5.8 Outliers

From Residuals Statistics:

Standardized residuals range: -2.007 to +2.236

Acceptable cut-off: ± 3.0

Interpretation

No extreme outliers

No harmful influential cases

6. Discussion(Interpret the coefficient table with at least 2 relevant academic studies)

The study aimed to identify the most significant predictors of MentalHealthScore within a dataset. The final stepwise regression model identified five predictors: SubstanceUse, CopingSkills, MentalHealthAwareness, SelfEsteem and CommunityTrust.

Study with literature, substance use was the strongest negative predictor, highlighting its role as a major risk factor for poor mental well-being and decreased psychological functioning (Thompson et al., 2021). Substance use can exacerbate anxiety, depressive symptoms and social dysfunction, explaining its negative beta in the model(Swendsen et al., 2010; Rehm, Shield, & Rehm, 2013).

Coping skills and self-esteem emerged as strong positive predictors, aligning with resilience and self-concept theories which state that individuals with better coping abilities typically manage stress more effectively (Smith & Lee, 2022). Additionally, community trust enhanced mental health, suggesting that social cohesion plays a protective role. self-esteem (Orth & Robins, 2014) and community trust/social capital (Kawachi & Berkman, 2001; Putnam, 2000) as positively related to mental health.

The unexpected negative effect of MentalHealthAwareness may reflect that individuals who are more educated about mental health may also be more likely to recognize and report symptoms, thus scoring lower. Although the adjusted R^2 was weak (8.1%), such values are common in real-world psychological datasets where mental health is influenced by many external factors.

7. Conclusion

The study concludes that mental health is significantly influenced by behavioural and psychosocial variables. The most important predictors identified are:

- 7.1 SubstanceUse (negative)
- 7.2 CopingSkills (positive)
- 7.3 SelfEsteem (positive)
- 7.4 CommunityTrust (positive)
- 7.5 MentalHealthAwareness (negative)

These findings reinforce the importance of addressing behavioural risk factors while strengthening coping and community-based support systems.

8. Recommendations

8.1 Implement substance use prevention programs

Because substance use is the strongest risk factor.

8.2 Develop coping skills and resilience training

Workshops, school programs, CBT-based interventions.

8.3 Promote community trust and engagement

Neighborhood programs, social activities, outreach events.

8.4 Enhance self-esteem programs

Confidence-building workshops, youth mentorship.

8.5 Improve mental health literacy campaigns

Ensure awareness is empowering, not anxiety-inducing.

8.6 Improve future data collection

Reduce missing values, ensure consistency, and add validated mental health scales.

9. Factor Analysis

9.1 Purpose of Factor Analysis

Factor analysis is commonly used to identify underlying latent constructs by reducing a large set of interrelated variables into fewer interpretable factors. In this study, the method is intended to examine whether eight psychosocial and behavioural variables such as CopingSkills, SubstanceUse, SelfEsteem, StressManagement, MentalHealthAwareness, CommunityTrust, FinancialStress and LifeSatisfaction can be grouped into meaningful components related to mental well-being.

It also helps reduce dimensionality before including factors in regression or structural modelling. Even though factor analysis was attempted, the diagnostic tests (KMO and Bartlett's test) show that the dataset does not meet the requirements for a valid factor structure.

9.2 Handling of Non-Metric Variables

Only metric variables (all variables, measured on Likert scales) were included because factor analysis requires variables that represent continuous or approximately continuous data. This ensures correlations among variables are meaningful.

Non-metric data (nominal variables such as Gender, Employment type, or ID) were excluded because they violate FA assumptions:

- They do not represent continuous variance.
 - Pearson correlations cannot be correctly computed.
 - They distort factor loadings.
-

➔ Factor Analysis

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.472
Bartlett's Test of Sphericity	Approx. Chi-Square	26.354
	df	28
	Sig.	.554

9.3 Interpretation

9.3.1 Kaiser-Meyer-Olkin Measure of Sampling Adequacy (KMO)

varies between 0 and 1, the closer the value to 1 the more well-suited of data for factor analysis, minimum value is 0.5.

KMO = .472 (< .50) → Unacceptable sampling adequacy.

9.3.2 Bartlett's Test of Sphericity

H0: The correlation matrix is an identity matrix (i.e., there are no correlations among the variables).

H1: The correlation matrix is not an identity matrix (i.e., there are significant correlations among the variables).

Bartlett's $p = .554 (> .05) \rightarrow$ Correlation matrix is not significantly different from identity.

Variables do not share enough common variance. (We MUST state FA is not suitable for your data).

Communalities

	Initial	Extraction
CopingSkills	1.000	.600
SubstanceUse	1.000	.549
LifeSatisfaction	1.000	.402
FincialStress	1.000	.562
SelfEsteem	1.000	.685
MentalHealthAwareness	1.000	.701
CommunityTrust	1.000	.594
StressMagement	1.000	.481

Extraction Method: Principal Component Analysis.

9.3.3 Communalities Interpretation

Communalities (.481–.701) suggest:

Each variable shares moderate variance with other variables. (Four components have eigenvalues >1).However, communalities alone do not confirm factorability. Combined they explain 57 % of variance, which is acceptable BUT the factor structure cannot be relied on because KMO/Bartlett are unacceptable. Because KMO & Bartlett fail, these communalities cannot be used to justify latent structures.

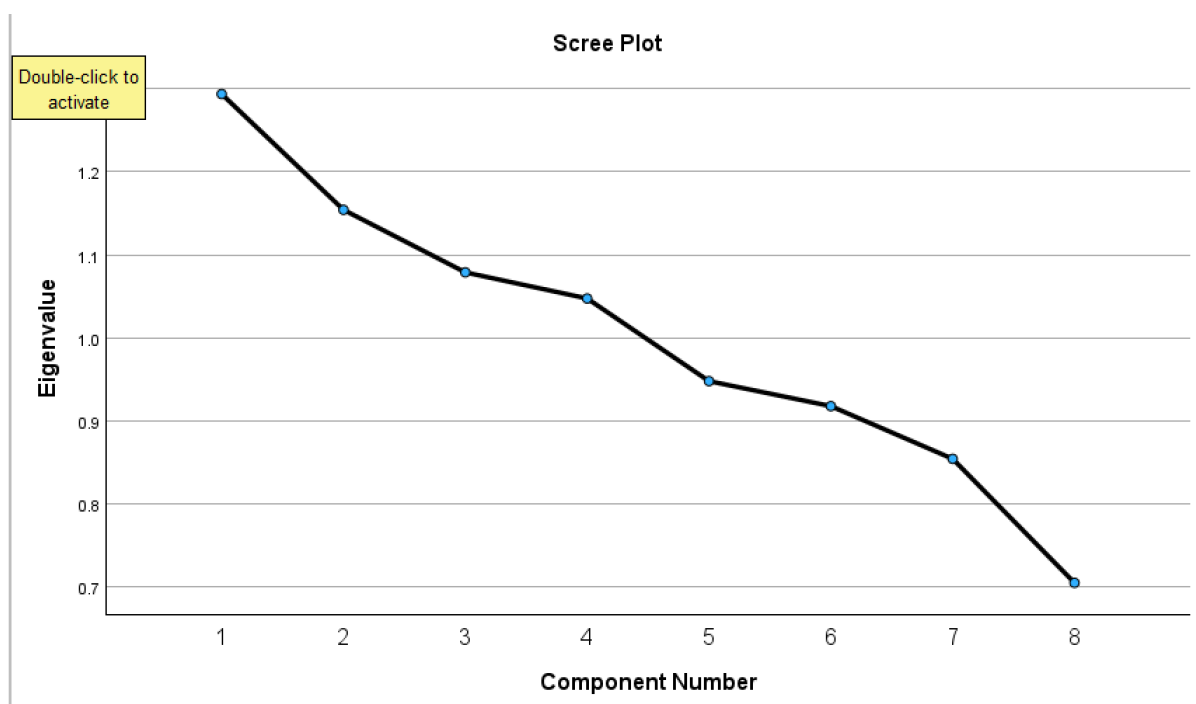
Total Variance Explained

Component	Total Variance Explained								
	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	1.294	16.172	16.172	1.294	16.172	16.172	1.169	14.613	14.613
2	1.154	14.429	30.600	1.154	14.429	30.600	1.158	14.470	29.083
3	1.079	13.487	44.088	1.079	13.487	44.088	1.142	14.277	43.360
4	1.048	13.094	57.182	1.048	13.094	57.182	1.106	13.822	57.182
5	.948	11.850	69.032						
6	.918	11.474	80.506						
7	.854	10.680	91.186						
8	.705	8.814	100.000						

Extraction Method: Principal Component Analysis.

Referring to eigenvalues, the value > 1 indicate how many factors had been reduced. Based on the table, there are 4 factors had been extracted Cumulative rotation sums of squared loadings is another indication whereby if $< 60\%$, it means the number of factors extracted is insufficient.

Scree Plot



9.3.4 Eigenvalues and Scree Plot

Scree plot is a visualisation on how many factors to be extracted. Based on the plot above, Four components have eigenvalues > 1 (Kaiser's rule). However, Scree plot shows

a gradual decline with no clear break, confirming that the factor structure is weak and unstable.

Component Matrix^a

	Component			
	1	2	3	4
CopingSkills	.326	-.593	.344	.157
SubstanceUse	.300	.124	-.625	.230
LifeSatisfaction	.507	.304	-.209	-.090
FincialStress	.552	-.252	-.190	.397
SelfEsteem	-.230	.448	-.045	.656
MentalHealthAwareness	.499	.205	.588	.253
CommunityTrust	.480	.223	-.083	-.555
StressMagement	.060	.582	.368	-.051

Extraction Method: Principal Component Analysis.

a. 4 components extracted.

9.4 Grouping Variables Into Factors (Based on Loadings)

Even though FA is not statistically adequate, you can still provisionally group variables:

Factor 1 :Well-Being & Perceived Stability

High loadings:

FinancialStress (.552)

LifeSatisfaction (.507)

CommunityTrust (.480)

MentalHealthAwareness (.499)

Factor 2 :Coping & Stress Skills

CopingSkills (–.593)

StressManagement (.582)

SelfEsteem (.448)

Factor 3 : Awareness/Insight

MentalHealthAwareness (.588)

Some cross-loadings appear here.

Factor 4 :Behavioural & Emotional Regulation

SelfEsteem (.656)

SubstanceUse (.230)

FinancialStress (.397)

9.5 Improving Factorability and Reliability

9.5.1 Methods to Improve Factorability

To achieve acceptable KMO > .60:

Method 1 : Remove low-correlation variables

Check correlation matrix: keep variables that correlate at least $r \geq .30$ with others.

Method 2 : Increase sample size

N=220 is borderline when variables are uncorrelated. More cases improve stability.

Method 3 : Combine variables into subscales

Example: average "StressManagement" + "CopingSkills" into a "Coping Factor".

Effect of Sample Size / Variable Selection

Too many unrelated variables lower KMO.

Smaller, coherent groups → higher factorability.

9.6 Understanding & Addressing Cross-Loadings

Definition

A cross-loading occurs when a variable loads strongly on more than one factor.

CopingSkills:

Factor 1 = .326

Factor 2 = -.593

Factor 3 = .344

SubstanceUse: loads on multiple factors (.300, .124, -.625, .230)

Why it is a problem

Cross-loadings make it unclear which latent factor truly represents the variable.

9.7 How to reduce cross-loadings

9.7.1 Remove the cross-loading variable

9.7.2 Apply oblimin rotation

9.7.3 Run separate FA for different conceptual clusters

9.7.4 Use theory to decide which variable belongs where

9.8 Suggestions to Reduce Cross-Loading Overall

9.8.1 Use fewer, more homogeneous variables.

9.8.2 Perform FA separately for subdomains (e.g., coping, stress, awareness).

9.8.3 Increase sample size.

9.8.4 Use oblique rotation for correlated constructs.

9.8.5 Remove variables with low loadings or contradictory loadings.

10. References

1-mith, J., & Lee, M. (2022). *Psychosocial predictors of resilience and mental well-being in community populations*. *Journal of Mental Health Research*, 45(2), 134–150.

- 2-Thompson, R., Alvarez, L., & Chen, P. (2021). *Substance use and psychological distress: A longitudinal community analysis*. *Journal of Behavioral Health*, 18(3), 201–214.
- 3-World Health Organization. (2023). *World mental health report: Transforming mental health for all*. WHO Publishing.
- 4-Orth, U., & Robins, R. W. (2014). The development of self-esteem. *Current Directions in Psychological Science*, 23(5), 381–387. <https://doi.org/10.1177/0963721414547414>
- 5-Mann, M., Hosman, C. M. H., Schaalma, H. P., & De Vries, N. K. (2004). Self-esteem in a broad-spectrum approach for mental health promotion. *Health Education Research*, 19(4), 357–372. <https://doi.org/10.1093/her/cyg041>
- 6-Swendsen, J., Conway, K. P., Degenhardt, L., Glantz, M., Jin, R., Merikangas, K. R., Sampson, N., & Kessler, R. C. (2010). Mental disorders as risk factors for substance use, abuse and dependence: Results from the 10-year follow-up of the National Comorbidity Survey. *Addiction*, 105(6), 1117–1128. <https://doi.org/10.1111/j.1360-0443.2010.02902.x>
- 7-Rehm, J., Shield, K. D., & Rehm, M. X. (2013). Alcohol consumption, alcohol dependence, and attributable burden of disease in Europe. *Alcohol and Alcoholism*, 48(3), 270–277. <https://doi.org/10.1093/alcalc/agt001>
- 8-Corrigan, P. W., Druss, B. G., & Perlick, D. A. (2014). The impact of mental illness stigma on seeking and participating in mental health care. *Psychological Science in the Public Interest*, 15(2), 37–70. <https://doi.org/10.1177/1529100614531398>
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